

Ali Alemami

Systems & Software Engineer

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Computer Science student (The Hashemite University '26) and 42 Amman Cadet specializing in low-level systems programming.
Solid foundation in **C** (POSIX APIs, multithreading, IPC, network I/O) and **C++** (Orthodox Canonical Form, STL, template metaprogramming).
Seeking Systems / Backend Software Engineering Internships or Junior roles.

TECHNICAL SKILLS

Programming Languages:	C (POSIX/C99), C++ (98/11/17), SQL (T-SQL), Bash
Systems & OS Concepts:	POSIX APIs, Multithreading (pthreads), Concurrency, Unix Signals & Pipes, File Descriptors, POSIX IPC, Memory Management
CS Fundamentals:	Data Structures & Algorithms, OOP & Orthodox Canonical Form, Computer Networks, Relational Schemas (3NF)
Tools & Infrastructure:	Linux, GDB, Valgrind, Docker, Docker Compose, Git, Makefiles, NGINX, SQL Server
Languages (Spoken):	Arabic (Native), English (Fluent / Professional)

EXPERIENCE

42 Amman — Software Engineering Cadet / Peer Developer 2025 – Present

Rigorous, peer-evaluated software engineering curriculum focused on pure C/C++, POSIX APIs, and systems programming.

- **Multithreading & Concurrency** (C — [philo](#)): Engineered a race-free simulation of Dijkstra's Dining Philosophers using pthreads and mutex locks; built async monitoring routines with microsecond-precision timers preventing thread starvation and deadlocks.
- **HTTP Web Server** (C++ — webserv, In Progress): Building a non-blocking HTTP/1.1 server using socket polling and I/O multiplexing, with stateful request parsing, NGINX-style config parsing, and CGI execution.
- **DevOps & Container Infrastructure** (inception, In Progress): Multi-container infrastructure with Docker Compose on Debian; isolated containers for NGINX (TLS/SSL), WordPress, and MariaDB with bind mounts and internal bridge networking.
- **Unix Process Pipelines & IPC** (C — [pipex](#)): Engineered a Unix pipeline engine replicating shell redirection (`< file1 cmd1 | cmd2 > file2`) using fork, pipe, dup2, and execve; implemented POSIX process synchronization, multi-fd stream redirection, PATH resolution, and error handling.
- **Algorithm Optimization & Dual-Stack Sorting** (C — [push_swap](#)): Built a dual-stack radix sort on array stacks with coordinate compression, bounding passes to $\log_2(N)$; added lookahead bit-checking in Stack B to cut redundant transfers, with heuristic decision trees for $N \leq 5$.
- **C++ Modules & Ford-Johnson Algorithm** (C++ — [cpps](#)): Completed 42 C++ Modules 00–09 (OOP, Orthodox Canonical Form, templates, STL); implemented Ford-Johnson Merge-Insert Sort optimizing comparisons via Jacobsthal sequence pairing.
- **Foundations & Algorithms**: [minitalk](#) (UNIX signal IPC), [get_next_line](#), [ft_printf](#), [libft](#).
- **Peer Review & Collaboration**: Conducted 50+ technical evaluations and collaborated in peer pair-programming sessions.

SYSTEMS, ALGORITHMS & DATABASE PROJECTS

- **Relational Database Schemas** (SQL Server — [Programming-Advices-Core](#)): Designed 5 normalized relational database schemas (3NF) with indexed foreign keys, integrity constraints, and optimized multi-table JOIN queries across varied business domains.
- **Bank & ATM Console Simulator** (C++ — [Bank-and-ATM-System-CPP](#)): Built a modular terminal-based banking application in C++ featuring role-based bitmask permissions, transaction auditing, and persistent file-based storage.
- **Algorithms & Problem Solving** (C++ — 225+ Solutions): Implemented solutions across 5 difficulty levels covering data structures, matrix operations, string parsing, and combinatorial algorithms.

EDUCATION

The Hashemite University — BSc in Computer Science

2022 – 2026 (Expected)